Spectrometer – Grating

Aim:

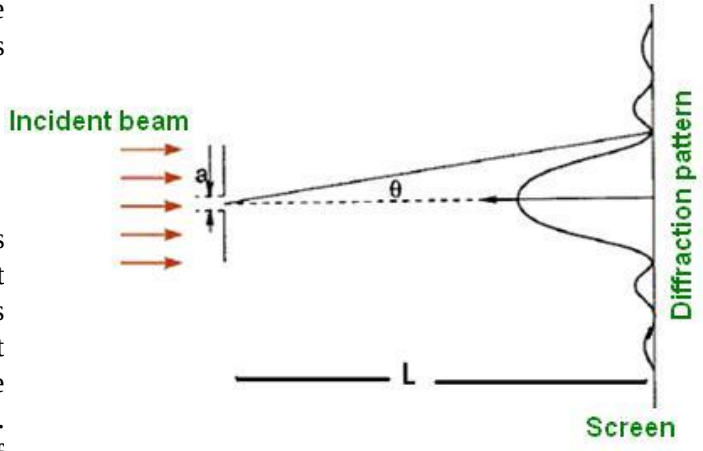
1. To determine the number of lines per millimetre of the grating using a sodium lamp.
2. To calculate the wavelength of the other prominent lines of Hydrogen.

Apparatus

Spectrometer, diffraction grating element, hydrogen lamp and sodium vapour lamp.

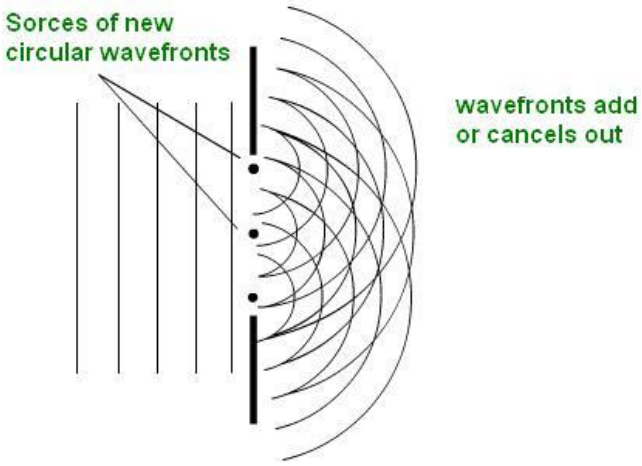
Theory:

When a wave train strikes an obstacle, the light ray will bend at the corners and edges of it, which causes the spreading of light waves into the geometrical shadow of the obstacle. This phenomenon is termed diffraction.



Single slit diffraction:

When waves pass through a gap about as wide as the wavelength, they spread out into the region beyond the gap. Huygens considered each point along a wave front to be the source of a secondary disturbance that forms a semi-circular wavelet. Diffraction is due to the superposition of such secondary wavelets. The secondary wavelets spread out and overlap, interfering with each other to form a pattern of maximum and minimum intensity. The pattern formed on a screen consists of a broad central band of light with dark bands on either side. The dark bands are caused when the light from the top half of the slit destructively interferes with the light from the bottom half.



Consider a slit of width ‘a’. Let at an angle θ, the path difference between the top and bottom of the slit is a wavelength. This causes destructive interference to occur because the path difference between the top and the middle of the slit is half of the wavelength. At this angle, all the light from the top half of the slit will get cancelled with the light from the bottom half to produce a dark band.

$$\Delta S = a\sin\theta \qquad (1)$$

Intensity minima will occur if this path length difference is an integer number of wavelengths.

$$a\sin\theta = n\lambda \qquad (2)$$

where,

n is the order of each minimum

λ is the wavelength,

a is the distance between the slits

θ is the angle at which destructive interference occurs.

Intensity is given by,

$$I = I_0 \frac{\sin^2\left(\frac{N\delta}{2}\right)}{\left(\frac{\delta}{2}\right)^2} \qquad (3)$$

where δ is the total phase angle, it can be related to the deviation angle, I_0 is the maximum intensity,

$$\delta = \frac{2\pi a\sin\theta}{\lambda} \qquad (4)$$

λ is the wavelength of the light and a is the slit width.

A diffraction grating is an optical component having a periodic structure which can split and diffract light into several beams travelling in different directions. This depends on the spacing of the grating and the wavelength of the incident light.
At normal incidence,

$$\sin\theta = Nn\lambda \qquad (5)$$

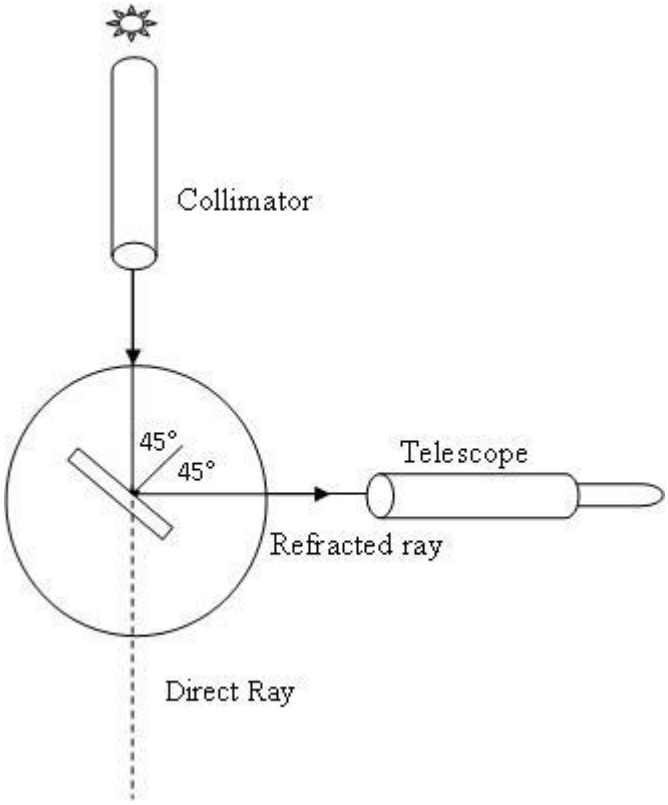
where,

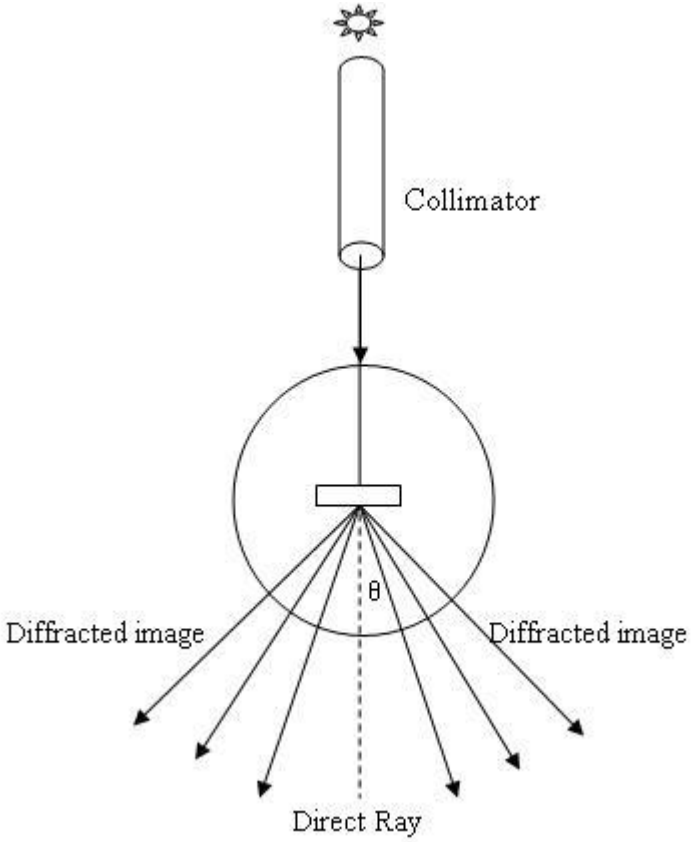
N is the number of lines per unit length of the grating

n is the order of the spectrum

λ is the wavelength of light.

θ is the diffraction angle.


Collimator
45°
45°
Telescope
Refracted ray
Direct Ray


Collimator
θ
Diffracted image
Diffracted image
Direct Ray

Procedure

1. The preliminary adjustments to the spectrometer are made. The grating is set for normal incidence. The slit is illuminated by a sodium vapour vamp.
2. The telescope is brought in a line with the collimator and the direct image of the slit is made to coincide with the vertical cross wire. The readings of one vernier are noted. The vernier table is firmly clamped.
3. Now, the telescope is rotated exactly through 90° and is fixed in this position. The grating is mounted vertically on the prism table with its ruled surface facing the collimator.
4. The vernier table is released and is slowly rotated till the reflected image coincides with the vertical cross wire. The levelling screws are adjusted so that the image is at the centre of the telescope's field of view.
5. The prism table is fixed and after making fine adjustments with the tangential screw, the readings of the vernier are noted. Now, the angle of incidence is 45°. The vernier table is then released and rotated exactly through 45° in the proper direction so that the surface of the grating becomes normal to the incident light. The vernier table is firmly clamped in this position.
6. The telescope is then released and is brought to observe the direct image. On either side of the direct image, the diffraction spectra are seen. The telescope is turned slowly towards the left so that the vertical cross wire coincides with the yellow lines of the first order.
7. The readings of the vernier are taken. The vertical cross wire is then made to coincide with the other lines on the left and the vernier readings are taken in each case. The telescope is then moved to the right and the reading of different lines is similarly taken.
8. The difference between the readings on the left and right on the same vernier is determined for each line. The mean value of this difference gives 2θ-twice the angle of diffraction. Thus, the angle of diffraction θ for each spectral line is determined. The wavelength of the yellow line is 580×10^{-9} m.
9. The number of lines per meter (N) of the grating is calculated. Using this value of N, the wavelengths of the other prominent lines in this spectrum are calculated.

To calculate the wavelength of different lines

1. Obtain the direct image.
2. Telescope is moved to make the cross-wire coincide with each line of the spectrum.
3. Note the readings on the verniers and calculate the diffraction angle.
4. Then calculate the wavelength of each colour.

Observations and Calculations

Standardization of equipment

Yellow, λ (nm)	Left		Right		Difference reading (2θ)		Mean θ	N=sinθ/nλ
	Ver 1	Ver 2	Ver 1	Ver 2	Ver 1	Ver 2		

For yellow light, λ = 580 nm

Determination of wavelength for prominent lines

Colour	Left		Right		Difference reading (2θ)		Mean θ	λ=sinθ/nN
	Ver 1	Ver 2	Ver 1	Ver 2	Ver 1	Ver 2		
Violet								
Blue								
Yellow								
Red								

Result:

The number of grating lines = _____
The wavelength of Violet = _____ nm
The wavelength of Blue = _____ nm
The wavelength of Yellow = _____ nm
The wavelength of Red = _____ nm

Discussion

Discuss the possible errors in the experiment.